Assignment 2
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1. Create a directory named `my_photos`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ mkdir my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ ls
my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$
```

2. Create a subfolder inside `my_photos` named `2024`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ ls
my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ cd my_photos/
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ mkdir 2024
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

3. Create an empty file inside `2024` named `vacation.jpg` (even though it's just a text placeholder).

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos/2024
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ ls
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ touch vacation.jpg
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ ls
vacation.jpg
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$
```

4. Create another empty file in `2024` named `family.png`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos/2024
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ ls
vacation.jpg
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ touch family.png
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ ls
family.png  vacation.jpg
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$
```

5. List only the files inside `2024` that end with `.png`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos/2024
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ ls
family.png  vacation.jpg
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ ls *.png
family.png
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$
```

6. Copy `family.png` to the `my_photos` directory.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ ls
family.png  vacation.jpg
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ cp family.png ~/Assignment/Assignment_2/my_photos/
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/2024$ cd ..
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024  family.png
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

7. Rename the copy of `family.png` in `my_photos` to `home.png`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024  family.png
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ mv family.png home.png
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024  home.png
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

8. Create a directory named `trash`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024  home.png
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ mkdir trash
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024  home.png  trash
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

9. Move `home.png` into the `trash` directory.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos/trash
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024  home.png  trash
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ mv home.png ~/Assignment/Assignment_2/my_photos/trash/
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ cd trash/
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/trash$ ls
home.png
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/trash$
```

10. Delete the `trash` directory and everything inside it.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos/trash$ cd ..
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024  trash
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ rm -r trash/
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

11. Create a file named `grocery_list.txt`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ touch grocery_list.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024  grocery_list.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

12. Add the word "Milk" to the `grocery_list.txt` file.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ echo "Milk" > grocery_list.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ cat grocery_list.txt
Milk
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

13. Add the word "Eggs" to the same file on a new line.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ cat grocery_list.txt
Milk
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ echo "Eggs" >> grocery_list.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ cat grocery_list.txt
Milk
Eggs
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

14. Display the contents of `grocery_list.txt` on your screen.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ cat grocery_list.txt
Milk
Eggs
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

15. Count how many lines are in your `grocery_list.txt`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ cat grocery_list.txt
Milk
Eggs
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ wc -l grocery_list.txt
2 grocery_list.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

16. Create a file named `hello.txt` with the text "Hello World".

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ echo "Hello World" > hello.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ ls
2024  grocery_list.txt  hello.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ cat hello.txt
Hello World
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

17. Overwrite `hello.txt` so it only says "Goodbye".

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ cat hello.txt
Hello World
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ echo "Goodbye" > hello.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ cat hello.txt
Goodbye
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

18. Search for the letter "G" in the `hello.txt` file.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ cat hello.txt
Goodbye
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$ grep "G" hello.txt
Goodbye
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/my_photos$
```

19. Create a directory named `scripts`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ ls
my_photos
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ mkdir scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ ls
my_photos  scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$
```

20. Move into the `scripts` directory using the `cd` command.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ ls
my_photos  scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ cd scripts/
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

21. Show the full path of the `scripts` directory you are currently in.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ ls
my_photos  scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ cd scripts/
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ pwd
/home/digvijay/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

22. Create an empty file named `test.py` inside `scripts`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ls
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ touch test.py
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ls
test.py
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

23. Move back to your home directory using a single shortcut.

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cd ~
digvijay@digvijay-virtualbox:~$
```

24. List all files in your home directory, including the hidden ones.

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cd ~
digvijay@digvijay-virtualbox:~$ ls -a
.                bda          list_to_delta  .secret                                  .vboxclient-display-svga-x11-tty2-control.pid  .vboxclient-seamless-tty2-service.pid
..               .cache       .local         snap                                     .vboxclient-display-svga-x11-tty2-service.pid  .vboxclient-vmsvga-session-tty2-control.pid
Assignment       .config      Music          .sudo_as_admin_successful                .vboxclient-draganddrop-tty2-control.pid       Videos
.bash_history    Desktop      Pictures       Templates                                .vboxclient-draganddrop-tty2-service.pid       workspace
.bash_logout     Documents    .profile       .vboxclient-clipboard-tty2-control.pid   .vboxclient-hostversion-tty2-control.pid       .xsession-errors
.bashrc          Downloads    Public         .vboxclient-clipboard-tty2-service.pid   .vboxclient-seamless-tty2-control.pid
digvijay@digvijay-virtualbox:~$
```

25. Use the `history` command to see the last 5 commands you typed.

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~$ history | tail -5
  429  ls -a
  430  clear
  431  history
  432  clear
  433  history | tail -5
digvijay@digvijay-virtualbox:~$
```

26. Use the `cal` command to see the calendar for the current month.

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~$ cal
     March 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31

digvijay@digvijay-virtualbox:~$
```

27. Find out how long your computer has been powered on (uptime).

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~$ uptime
 23:35:11 up 43 min,  1 user,  load average: 0.05, 0.04, 0.01
digvijay@digvijay-virtualbox:~$
```

28. Clear your terminal screen so it is empty again.

```
digvijay@digvijay-virtualbox: ~
digvijay@digvijay-virtualbox:~$ uptime
 23:35:11 up 43 min,  1 user,  load average: 0.05, 0.04, 0.01
digvijay@digvijay-virtualbox:~$ clear
```

29. Create a file named `sorting.txt` with the numbers 3, 1, and 2 on separate lines.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ ls
my_photos  scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ echo -e "3\n1\n2" > sorting.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ ls
my_photos  scripts  sorting.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ cat sorting.txt
3
1
2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$
```

30. Display the numbers from `sorting.txt` in alphabetical/numerical order.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ cat sorting.txt
3
1
2
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$ sort sorting.txt
1
2
3
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2$
```

Shell scripts

1. Write a script that asks for a user's name and prints a "Hello" message.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_1.sh
#!/bin/bash

read -p "Enter your name: " name
echo "Hello $name"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_1.sh
Enter your name: DS
Hello DS
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

2. Create a script that takes three separate strings as input and displays them in reverse order.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_2.sh
#!/bin/bash

read -p "Enter first string: " str1
read -p "Enter second string: " str2
read -p "Enter third string: " str3

echo "Reverse order:"
echo "$str3 $str2 $str1"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_2.sh
Enter first string: Sawant
Enter second string: Vikas
Enter third string: Digvijay
Reverse order:
Digvijay Vikas Sawant
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

3. Write a script that asks for a user's age and calculates how many years are left until they turn 100.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_3.sh
#!/bin/bash

read -p "Enter your age: " age
years=$((100-age))
echo "$years years left until you turn 100"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_3.sh
Enter your age: 22
78 years left until you turn 100
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

4. Create a script that prompts for a filename and then creates that file using `touch`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_4.sh
#!/bin/bash

read -p "Enter file name: " fileName
touch $fileName
echo "$fileName file is created"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_4.sh
Enter file name: Text1
Text1 file is created
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

5. Write a script that asks for a directory name and lists all files inside that directory.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_5.sh
#!/bin/bash

read -p "Enter directory name: " dir
ls $dir
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_5.sh
Enter directory name: Sample
S1.txt
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

6. Write a script to take two numbers as input and display their sum.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_6.sh
#!/bin/bash

read -p "Enter first number: " num1
read -p "Enter second number: " num2

echo "Sum of $num1 and $num2: $((num1+num2))"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_6.sh
Enter first number: 10
Enter second number: 20
Sum of 10 and 20: 30
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

7. Create a script that calculates the area of a rectangle (Length × Width) based on user input.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_7.sh
#!/bin/bash

read -p "Enter length of rectangle: " len
read -p "Enter width of rectangle: " wid

echo "Area of rectangle: $((len*wid))"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_7.sh
Enter length of rectangle: 10
Enter width of rectangle: 5
Area of rectangle: 50
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

8. Write a script that takes a temperature in Celsius and converts it to Fahrenheit.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_8.sh
#!/bin/bash

read -p "Enter temperature in Celsius: " temp

#fahr=$(echo "scale=2; ($temp * 9/5) + 32" | bc)
echo "Temperature in Fahrenhiet: $(echo "scale=2; ($temp * 9/5) + 32" | bc)"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_8.sh
Enter temperature in Celsius: 32
Temperature in Fahrenhiet: 89.60
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

9. Create a script that performs all four basic operations (+, -, *, /) on two provided numbers.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_9.sh
#!/bin/bash

read -p "Enter first number: " num1
read -p "Enter second number: " num2

echo "Addition: $((num1+num2))"
echo "Substraction: $((num1-num2))"
echo "Multiplication: $((num1*num2))"
echo "Division: $(echo "scale=2; ($num1/$num2)"| bc)"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_9.sh
Enter first number: 10
Enter second number: 5
Addition: 15
Substraction: 5
Multiplication: 50
Division: 2.00
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

10. Write a script to calculate the square and cube of a number entered by the user.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_10.sh
#!/bin/bash

read -p "Enter the number: " num

echo "Square: $((num*num))"
echo "Cube: $((num*num*num))"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_10.sh
Enter the number: 5
Square: 25
Cube: 125
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

11. Create a script that finds the remainder (modulus) when one number is divided by another.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_11.sh
#!/bin/bash

read -p "Enter the dividend: " dividend
read -p "Enter the divisor: " divisor

remainder=$((dividend%divisor))

echo "Remainder when $dividend is divided by $divisor = $remainder"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_11.sh
Enter the dividend: 10
Enter the divisor: 5
Remainder when 10 is divided by 5 = 0
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

12. Write a script that takes a total number of minutes and converts it into "X hours and Y minutes" format.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_12.sh
#!/bin/bash

read -p "Enter the total timing in minutes: " total_minutes

hours=$((total_minutes/60))
rem_min=$((total_minutes%60))

echo "The given time $total_minutes min is = $hours hr and $rem_min min"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_12.sh
Enter the total timing in minutes: 67
The given time 67 min is = 1 hr and 7 min
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

13. Write a script that checks if a number entered by the user is positive, negative, or zero.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_13.sh
#!/bin/bash

read -p "Enter the number: " num

if [ $num -gt 0 ]
then
        echo "The number is positive"
elif [ $num -lt 0 ]
then
        echo "The number is negative"
elif [ $num -eq 0 ]
then
        echo "The number is zero"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_13.sh
Enter the number: -10
The number is negative
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

14. Create a script that asks for a number and determines if it is even or odd.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_14.sh
#!/bin/bash

read -p "Enter the number: " num

if (( $num%2==0 ))
then
        echo "The number is Even number"
else
        echo "The numner is odd number"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_14.sh
Enter the number: 25
The numner is odd number
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

15. Write a script that checks if a user is eligible to vote (age 18 or older).

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_15.sh
#!/bin/bash

read -p "Enter the user age: " age

if (( $age >= 18 ))
then
        echo "Then the user is eligible for voting"
else
        echo "Then the user is not eligible for voting"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_15.sh
Enter the user age: 22
Then the user is eligible for voting
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

16. Create a script that compares two numbers and prints which one is larger.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_16.sh
#!/bin/bash

read -p "Enter the first number: " num1
read -p "Enter the second number: " num2

if (( $num1 > num2 ))
then
        echo "Then the larger number is first number: $num1"
else
        echo "Then the larger number is second number: $num2"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_16.sh
Enter the first number: 20
Enter the second number: 23
Then the larger number is second number: 23
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

17. Write a "Pass/Fail" script: if a score is 50 or above, print "Pass", otherwise "Fail".

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_17.sh
#!/bin/bash

read -p "Enter the marks obtained by student: " marks

if [ "$marks" -ge 50 ]; then
        echo 'Student is Passed'
elif [ "$marks" -lt 50 ]; then
        echo 'Student is Failed'
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_17.sh
Enter the marks obtained by student: 55
Student is Passed
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

18. Create a script that checks if a specific file (input by the user) exists on the system.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_18.sh
#!/bin/bash

read -p "Enter file name: " filename

if [ -f "$filename" ]; then
        echo "File exists on system"
else
        echo "File doesn't on system"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_18.sh
Enter file name: script_10.sh
File exists on system
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

19. Write a script that asks for a password and prints "Access Granted" if it matches "Secret123", and "Access Denied" otherwise.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_19.sh
#!/bin/bash

read -p "Enter the password: " password

if [ $password = "Secret123" ]; then
        echo "Access Granted"
else
        echo "Access Denied"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_19.sh
Enter the password: Secret123
Access Granted
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

20. Create a script that checks if a given year is a leap year.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_20.sh
#!/bin/bash

read -p "Enter the year: " year

if (( ($year%4 ==0 && $year%100 != 0) || ($year%400 == 0) )); then
        echo "This $year is leap year"
else
        echo "This $year is not leap year"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_20.sh
Enter the year: 2026
This 2026 is not leap year
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

21. Write a script that takes a numerical grade (0-100) and prints the letter grade (A, B, C, D, or F) using `elif`.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_21.sh
#!/bin/bash

read -p "Enter the marks obtained by student: " marks

if [ "$marks" -ge 90 ]; then
        echo 'Grade obtained = A'
elif [ "$marks" -gt 80 ]; then
        echo 'Grade obtained = B'
elif [ "$marks" -gt 65 ]; then
        echo 'Grade obtained = C'
elif [ "$marks" -gt 55 ]; then
        echo 'Grade obtained = D'
elif [ "$marks" -gt 40 ]; then
        echo 'Grade obtained = E'
elif [ "$marks" -lt 40 ]; then
        echo 'Grade obtained = F'
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_21.sh
Enter the marks obtained by student: 85
Grade obtained = B
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

22. Create a script that asks for a username; if the username is "admin", it asks for a password; otherwise, it exits.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_22.sh
#!/bin/bash

read -p "Enter the username: " username

if [ $username = "admin" ]; then
        read -p "Enter the Password: " password
        echo "Password Entered"
else
        echo "Access Denied"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_22.sh
Enter the username: admin
Enter the Password: 1234
Password Entered
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

23. Write a script that checks if a string entered by the user is empty or not.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_23.sh
#!/bin/bash

read -p "Enter the string: " str

if [ -z "$str" ]; then
        echo "String is Empty"
else
        echo "String is not Empty"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_23.sh
Enter the string:
String is Empty
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_23.sh
Enter the string: abcd
String is not Empty
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

24. Create a simple calculator where the user enters two numbers and an operator (+ or -) to get the result.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_24.sh
#Simple calculator
#!/bin/bash

read -p "Enter the first number: " num1
read -p "Enter the second number: " num2
read -p "Enter the operator (+,-): " op

#check each condition using if/elif
if [ "$op" == "+" ]; then
        result=$(echo "$num1 + $num2" | bc)
elif [ "$op" == "-" ]; then
        result=$(echo "$num1 - $num2" | bc)
else
        echo "Invalid operator selected."
        exit 1
fi

echo "Result = $result"
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_24.sh
Enter the first number: 10
Enter the second number: 5
Enter the operator (+,-): -
Result = 5
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```

25. Write a script that checks if a user-provided path is a file or a directory.

```
digvijay@digvijay-virtualbox: ~/Assignment/Assignment_2/scripts
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ cat script_25.sh
#!/bin/bash

read -p "Enter the path: " path

if [ -f "$path" ]; then
        echo "This is a file"
elif [ -d "$path" ]; then
        echo "This is the directory"
else
        echo "This path does not exists"
fi
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$ ./script_25.sh
Enter the path: Sample
This is the directory
digvijay@digvijay-virtualbox:~/Assignment/Assignment_2/scripts$
```